Week 9 - Clouds and Precipitation: Practice Questions

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS**: selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY**: expected independent practice after the session.
- **EXTRA CHALLENGE**: optional problems for students who want additional depth or a harder application.

Attempt the questions before looking at the worked answers. Show equations, substitutions and units for numerical work, and use sketches where they help. You are **not** expected to complete every Extra Challenge problem. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

Useful values used in several questions: liquid-water density $\rho_w = 1000\ \mathrm{kg\,m^{-3}}$, latent heat of vaporisation $L_v \approx 2.5 \times 10^6\ \mathrm{J\,kg^{-1}}$. Other constants are supplied where needed.

YOU SHOULD TRY Question 1: Classify clouds from observable features

For each observation, identify the most appropriate cloud type and give the feature that distinguishes it.

1. Thin, fibrous ice cloud at about 9 km.
2. A low, uniform grey layer with a base near 500 m and little vertical structure.
3. A deep cloud with a dark base, strong vertical towers and an anvil, producing heavy showers.
4. A widespread, thick layered cloud producing persistent rain ahead of a warm front.

EXTRA CHALLENGE Question 2: Why pure-water droplets are difficult to form: the Kelvin effect

For a spherical pure-water droplet, a simplified Kelvin relation for the equilibrium saturation ratio is

$$\mathcal{S}_{eq}(r) = \exp\left(\frac{A_K}{r}\right), \qquad A_K = \frac{2\sigma_w M_w}{\rho_w R^* T}.$$

Use $\sigma_w = 0.075\ \mathrm{N\,m^{-1}}$, $M_w = 0.018\ \mathrm{kg\,mol^{-1}}$, $R^* = 8.314\ \mathrm{J\,mol^{-1}\,K^{-1}}$, $T = 283$ K and $\rho_w = 1000\ \mathrm{kg\,m^{-3}}$.

1. Calculate A_K in metres.
2. Calculate the equilibrium supersaturation $s = 100(\mathcal{S}_{eq} - 1)$ for pure-water droplets of radius 0.01, 0.10 and 1.0 μm.
3. A cloud parcel reaches a maximum supersaturation of 0.5%. Which of these pure-water droplets could persist/grow according to this calculation?
4. Explain why soluble hygroscopic aerosol particles can activate as cloud-condensation nuclei at much smaller atmospheric supersaturations than equally small pure-water embryos.

IN CLASS Question 3: The same cloud water divided among different numbers of droplets

Cloud A contains $N_A = 100$ cm^{-3} droplets of mean radius $r_A = 10$ μm. Cloud B has the same liquid-water content but $N_B = 800$ cm^{-3}. Assume, for this exercise, that all droplets within a cloud have the same radius.

1. Explain why fixed liquid-water content implies $Nr^3 = \text{constant}$.
2. Calculate the mean radius r_B.
3. Define the area-related quantity $M_2 = Nr^2$. Calculate the ratio $M_{2,B}/M_{2,A}$.
4. Cloud-droplet extinction is approximately proportional to an area measure such as M_2. Which cloud therefore has the larger extinction coefficient?
5. Which cloud is likely to be more favourable for the early onset of collision–coalescence? Explain your reasoning.

YOU SHOULD TRY Question 4: Growth by condensation: why it cannot make raindrops quickly

For a fixed cloud environment, use the simplified condensational-growth law

$$r^2 - r_0^2 = Kt,$$

with r in μm, t in seconds and $K = 0.125$ $\mu\text{m}^2\,\text{s}^{-1}$.

1. How long does a droplet take to grow from 5 to 10 μm radius?
2. How long does the next doubling, from 10 to 20 μm, take?
3. If the same value of K unrealistically remained valid, how long would growth from 20 μm to a 1.0 mm radius raindrop take?
4. Explain what the calculation tells you about the need for another precipitation-growth mechanism.

YOU SHOULD TRY Question 5: Derive and use Stokes terminal fall speed

For a small spherical droplet, take the upward Stokes drag to be

$$F_D = 6\pi\eta r v_t,$$

and the downward net gravitational force to be

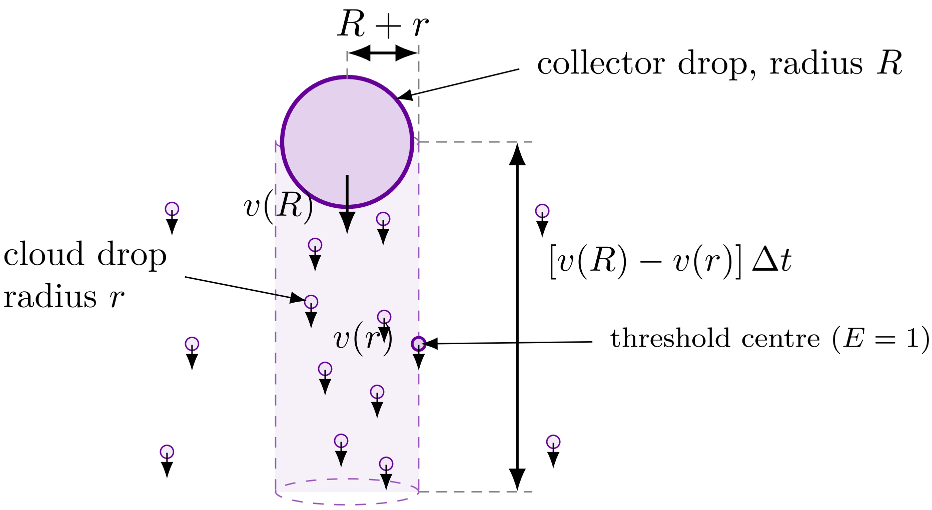
$$F_g = \frac{4}{3}\pi r^3(\rho_w - \rho_a)g.$$

Use $\eta = 1.7 \times 10^{-5}$ Pa s, $\rho_w = 1000$ kg m^{-3}, $\rho_a = 1.2$ kg m^{-3} and $g = 9.81$ m s^{-2}.

1. Set $F_D = F_g$ and derive an expression for v_t.
2. Calculate v_t for radii 10, 20 and 30 μm.
3. By what factor does the fall speed change when radius triples from 10 to 30 μm?
4. Why is this size dependence important for collision–coalescence?

IN CLASS Question 6: Collision–coalescence growth and spectral broadening

Consider a collector drop of radius R falling through a population of smaller cloud droplets of radius r and number concentration N. The collector falls at speed $v(R)$ and the small droplets at speed $v(r)$. Assume collision efficiency $E = 1$ for this idealised derivation.



A small-drop centre exactly on the cylinder wall passes the collector at centre-to-centre separation $R + r$: the geometrical threshold for contact when $E = 1$.

The geometrical sweep-out rate is

$$\frac{dV_{\text{swept}}}{dt} = \pi(R + r)^2\,[v(R) - v(r)].$$

Let

$$c = \frac{4}{3}\pi r^3$$

be the volume of one small cloud droplet. Hence the collector-volume growth rate is

$$\frac{dV}{dt} = \pi(R + r)^2\,[v(R) - v(r)]\,Nc.$$

1. Assume that the collector is much larger and falls much faster than the small droplets, so that $R + r \simeq R$ and $v(R) - v(r) \simeq v(R)$. If $v(R) = aR^b$, show that

$$\frac{dV}{dt} \simeq \pi R^2 aR^b Nc.$$

2. The collector volume is $V = (4/3)\pi R^3$. Use this to show that

$$\frac{dR}{dt} = \frac{aNc}{4}R^b.$$

3. For $b = 0.5$, solve the differential equation for a collector of initial radius R_0 and show that

$$R(t) = \left[R_0^{1/2} + \frac{aNc}{8}t\right]^2.$$

4. Use the form of this solution to explain why two collectors with different initial radii move farther apart in radius as time increases, and hence why collision–coalescence tends to broaden a droplet-size distribution.

IN CLASS Question 7: The Wegener–Bergeron–Findeisen process quantitatively

At $-10°$C, suppose saturation vapour pressure is 2.86 hPa over liquid water and 2.60 hPa over ice. A mixed-phase cloud is exactly saturated with respect to liquid water.

1. What is the actual vapour pressure?
2. Calculate relative humidity with respect to ice.
3. Is the air saturated, subsaturated or supersaturated with respect to ice?
4. State the resulting direction of net water transfer between supercooled droplets and ice crystals.

YOU SHOULD TRY Question 8: How much water is in a cloud?

Approximate a cloud as a cylinder of diameter 1.0 km and depth 2.0 km. Its mean liquid-water content is $0.40\ \mathrm{g\,m^{-3}}$.

1. Calculate the cloud volume.
2. Calculate the total mass of liquid water.
3. Estimate the latent heat released in condensing this liquid water from vapour.
4. If all the liquid water fell uniformly onto the circular area directly beneath the cloud, what rainfall depth would it produce?

EXTRA CHALLENGE Question 9: Choose the precipitation observing system

A catchment has one well-maintained rain gauge and is also covered by a weather radar. For each task, choose the more useful primary measurement and justify your choice.

1. Determine the accumulated rainfall at the exact gauge location over 24 hours.
2. Track the motion of a narrow band of intense rain across the whole catchment during the next two hours.
3. Estimate whether rainfall varies strongly from one side of the catchment to the other.

Then name one important source of error for the gauge and two for radar rainfall estimates.

EXTRA CHALLENGE Question 10: Why large hail requires strong updraughts

A hailstone has a downward terminal speed of $25\ \mathrm{m\,s^{-1}}$ inside a storm. Consider two updraughts: Storm A has $w = 15\ \mathrm{m\,s^{-1}}$ and Storm B has $w = 35\ \mathrm{m\,s^{-1}}$.

1. In which storm can the hailstone initially be carried upward relative to the ground?
2. What is its approximate vertical ground-relative velocity in each storm if terminal speed and updraught simply oppose one another?
3. Explain why strong updraughts allow larger hail to grow by repeated exposure to supercooled liquid water.